# 03 CONCRETE PIERS
# CONTAINER FOUNDATION

*Six level concrete piers for a 20-ft high-cube container.*

| LEVEL | CREW | WHERE | STAGE NOW |
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| ■■□ Some skill or supervision | 2–3 | To be decided (flat ground with truck access) | Siting, layout, digging |

A 20-ft high-cube shipping container is about 20 ft long, 8 ft wide and 9 ft 6 in tall, and weighs roughly 5,000 lb empty. It carries its weight on four steel corner blocks (corner castings). So it doesn't need a slab: it needs piers that are exactly level and sit on firm ground.

## FIRST STEPS *today*

1. Choose the spot with Yusuf: flat, reachable by the delivery truck, out of the dome's way and out of the neighbors' sightlines.
2. Clear and rough-level the area. String the container rectangle (19 ft 10½ in × 8 ft) with batter boards. Check that both diagonals measure the same.
3. Mark six pier centers: each corner pier about 4 in in from both outside edges (under the corner block), plus one at the middle of each long side. Measure the actual container if you can.
4. Dig the six holes about 18–24 in wide, to firm soil or rock (see Rocky ground). Add and tamp 4–6 in of gravel where there's soil at the bottom.
5. Set the tubes a little taller than needed and brace them plumb with 2×4s. Then mark and cut all six to one height using the laser steps below.

## BEFORE YOU START

- Call 811 (DigAlert) before digging. Check the water-line route with Yusuf.
- Delivery: a tilt-bed truck needs a long, straight, clear approach and overhead clearance. Confirm access up the canyon road before ordering, or plan for a crane.
- A 20-ft container is 160 sq ft, larger than the size California typically exempts from permits for accessory structures (commonly 120 sq ft). Yusuf checks with the County first.
- Wet concrete is caustic: gloves, long sleeves, glasses.

## THE BASICS

**How many piers** — Four corner piers is the minimum. Six (four corners + two at the middle of each long side) is better for a container that will be lived in or loaded, or on softer soil.

**The "circle things"** — Cardboard concrete form tubes (Sonotube is the common brand). Use 12 in diameter, 48 in length.

**How deep** — There's no frost here, so depth is about firm ground, not a number: dig until you hit undisturbed soil, usually 24–36 in, deeper on the downhill side of a slope, with 4–6 in of compacted gravel in the bottom. Hit rock after a few inches? That's often good news: see Rocky ground below.

**How high** — Let the tube stand 4–6 in above ground. That keeps the steel off wet soil, lets air move underneath and makes rodents less comfortable.

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| **Level is everything** | All six pier tops must sit at the same height, within about ⅛ in. A bubble level can't do that across 20 ft. You need a tool that throws one fixed height across the whole site: a rotary laser (digital) or a water level (traditional). Step-by-step below. |

## HOW-TO NOTES

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| **Rocky ground: first, read the rock** | Strike it with a digging bar (a heavy steel pry bar). Solid bedrock rings and doesn't move. A loose rock or buried boulder thuds, shifts, or rocks when you lever it. |
| **If it's solid bedrock** | Stop digging: rock is a better base than any soil. Scrape it clean, chip a roughly flat seat (a cold chisel or rented rotary hammer), drill 2–3 holes 4–6 in into the rock with a hammer drill, and glue rebar into them with anchoring epoxy. Set the tube on the rock around the rebar and pour. The pier is now pinned to the mountain. |
| **If it's loose rock** | Lever the rocks out with the bar and keep going to firm ground. A wider, messier hole is fine: concrete or tamped gravel fills the gaps around the tube. Save every rock for the front wall's plinth. |
| **If a boulder won't budge** | If it doesn't move when levered, treat it like bedrock. If it's in the way of a middle pier, slide that pier a few inches along the long side. Corner piers must stay under the corner blocks. |
| **Last resort: pad footings** | If a spot is a jumble of rock that can't be cleared, pour a wide pad instead of a deep pier: a 24 × 24 in, 12 in thick reinforced concrete pad on cleared, compacted ground or rock, with the tube or a short pier on top. In a frost-free climate this works when the ground under it is firm. |
| **Mixed results** | Some piers on rock, some on soil is fine as long as every one bears on something firm. Widen the bottom of the soil holes (to about 24 in) so they can't settle while the rock ones can't move. |
| **Leveling all six, step by step (laser)** | **Kit:** a self-leveling rotary laser on a tripod, a laser detector (receiver), and a grade rod (a tall measuring pole the detector clamps onto). Rent it or buy a kit (roughly $200–400).<br>**1.** Set the laser on its tripod near the middle, where it can see all six tubes. Switch it on: it levels itself and spins a beam at one exact height all around the site.<br>**2.** Find the tube where the ground is highest. Pencil a mark on it 4–6 in above the ground. This mark is the top of every pier.<br>**3.** Stand the grade rod against that tube with its foot exactly on the mark. Slide the detector up or down the rod until it gives the steady "on-grade" tone. Lock the detector there. Don't touch it again.<br>**4.** Carry the rod to the next tube. Hold it against the tube and slide the whole rod up or down until you hear the same steady tone. Mark the tube at the rod's foot. Repeat on all six. Every mark is now at the same height.<br>**5.** Cut each tube at its mark, or drive a screw through the tube at the mark as a fill line. Re-brace.<br>**6.** Check all six again before pouring, and once more after troweling, while the concrete is still wet. |
| **The no-battery version** | A water level does the same thing: a long clear hose filled with water, no bubbles. Water settles at the same height at both ends. Hold one end at the mark on the first tube with the water line on the mark; carry the other end to each tube and mark where the water settles. |
| **Concrete math** | A 12 in tube holds about 0.8 cu ft per foot of height. A 3-ft pier is about 2.4 cu ft. An 80-lb bag of concrete makes about 0.6 cu ft, so each pier takes about 4 bags. Six piers: about 24 bags, so buy 26–28. |
| **Steel** | Two or three pieces of #4 (½ in) rebar standing in each pier. If the container will be tied down (wind, earthquakes), embed a steel plate with anchors in the top of each corner pier so the corner blocks can be welded or bolted to it. |

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| **Curing** | Keep the concrete damp and covered. Wait at least 7 days before the container is set down. |

## THE PLAN *in order*

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| 01 | **Decide** | Purpose of the container, spot, truck access, tie-down, permit. |
| 02 | **Layout** | Batter boards, string lines, diagonal check, pier marks. |
| 03 | **Dig + gravel** | Six holes to firm soil or rock; rocky-ground plan as needed. |
| 04 | **Tubes + level** | Set, brace, mark all six with the laser, cut, recheck. |
| 05 | **Pour** | Rebar in, concrete mixed in a tub or wheelbarrow, poured, rodded to remove air, tops troweled; plates set if used. |
| 06 | **Cure + deliver** | 7+ days, then delivery. |

## TOOLS & MATERIALS

- Post-hole digger or rented auger
- 6 × 12 in × 48 in form tubes
- About 26–28 × 80-lb concrete bags
- 12 pieces #4 rebar, ~30 in
- Gravel + hand tamper
- Rotary laser + detector + grade rod (rent or buy)
- Water level (clear hose) as backup
- Digging bar
- Hammer drill + rebar epoxy (if bedrock)
- String, stakes, batter boards, tape measure
- 2×4s and screws for bracing
- Hand saw for the tubes
- Mixing tub, hoe, wheelbarrow, water

## FOR YUSUF TO DECIDE

- ? What the container is for
- ? Where it goes
- ? Truck access or crane
- ? Tie-down plates: yes or no
- ? Permit route

**Resources:** DigAlert: call 811 (digalert.org) · LA County Public Works permits (EPIC-LA) (epicla.lacounty.gov)